

Elementary matrices

Video 6.1.1

The elementary matrices are a special class of extremely useful matrices.

Definition ('Elementary matrix').

An **elementary matrix** is any matrix which is obtained from an identity matrix by applying a single elementary row operation.

There are three types of elementary row operations, so there will be three types of elementary matrix.

Examples of the 3 types

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{R_1 \rightarrow R_1 - 5R_3} \begin{bmatrix} 1 & 0 & -5 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \xrightarrow{R_2 \leftrightarrow R_4} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \xrightarrow{R_2 \rightarrow 7R_2} \begin{bmatrix} 1 & 0 \\ 0 & 7 \end{bmatrix}$$

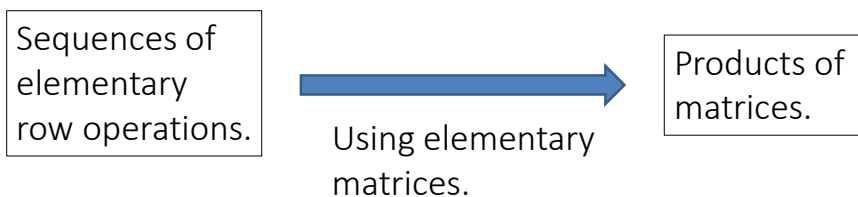
For example,
this is called the
elementary 3×3
matrix corresponding
to the elementary
row operation
 $R_1 \rightarrow R_1 - 5R_3$

Why are elementary matrices useful?

Video 6.1.2

We will see that the main reason they are useful is that when you multiply on the left with them they *perform* elementary row operations for you.

The idea:



Here is the **Fundamental Fact** about elementary matrices:

Let A be some $m \times n$ matrix.

Let B be the $m \times n$ matrix which results from applying some elementary row operation to A .

Let E be the result of applying the *same* row operation to I_m .

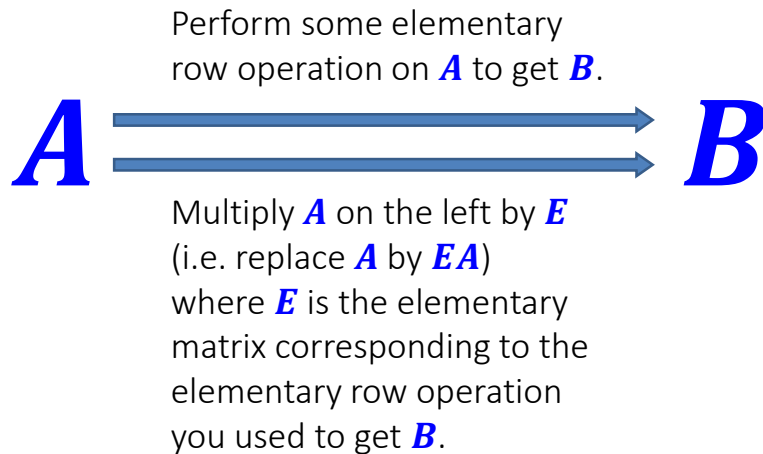
Then:

$$B = EA.$$

Comment: Note that this says that to build E you start by writing down the identity matrix I_m which fits A on its left and then apply the operation to that.

Diagram

The following pair of operations have *the same effect*:



Example

We'll illustrate this fundamental fact with the example where the starting matrix is

$$A = \begin{bmatrix} 1 & 0 & 2 & 3 \\ 2 & -1 & 3 & 6 \\ 1 & 4 & 4 & 0 \end{bmatrix}$$

and the involved elementary row operation is
 $R_3 \rightarrow R_3 - R_1$.

The fundamental fact says that two different operations corresponding to this elementary row operation have the same effect on A .

Example (continued)

Perform the elementary row operation $R_3 \rightarrow R_3 - R_1$

$$\begin{bmatrix} 1 & 0 & 2 & 3 \\ 2 & -1 & 3 & 6 \\ 1 & 4 & 4 & 0 \end{bmatrix} \xrightarrow{\quad} \begin{bmatrix} 1 & 0 & 2 & 3 \\ 2 & -1 & 3 & 6 \\ 0 & 4 & 2 & -3 \end{bmatrix}$$

Multiply on the left by the elementary matrix which corresponds to this operation, which is:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{R_3 \rightarrow R_3 - R_1} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} \quad \text{EA: } \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 2 & 3 \\ 2 & -1 & 3 & 6 \\ 1 & 4 & 4 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 2 & 3 \\ 2 & -1 & 3 & 6 \\ 0 & 4 & 2 & -3 \end{bmatrix}$$

Quick test of understanding:

Video 6.1.3

Let A be some 5×100 matrix.

$$\text{Let } E_1 = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \text{ and let } E_2 = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

$R_2 \leftrightarrow R_4$

$R_2 \rightarrow R_2 + R_4$

They are elementary matrices!

Apply the **Fundamental Fact** to describe how to obtain the matrix $E_1 E_2 A$ from A using some elementary row operations.

Answer:

* Need to first check that they are indeed elementary row operations

Answer:

We start with A .

To use the **Fundamental Fact** to understand how to get the matrix $E_1 E_2 A$ from A we must calculate the second product first. i.e. we will bracket as follows:

$$E_1(E_2 A)$$

First we work out how to get this part from A .

Then we work out how to get the final expression from what we found for $E_2 A$.

Answer:

So we must calculate the second product first. i.e. bracket as follows: $E_1(E_2A)$

Call this matrix B .
The **Fundamental Fact** says that B is obtained from A by replacing R_2 by $R_2 + R_4$

Now we must understand E_1B . The **Fundamental Fact** says that E_1B is obtained from B by the row operation $R_2 \leftrightarrow R_4$.

Putting these two deductions together we conclude:

The matrix E_1E_2A is obtained from A by

1. First applying the row operation $R_2 \rightarrow R_2 + R_4$
2. Then applying the row operation $R_2 \leftrightarrow R_4$.

□

Quick test of understanding:**Video 6.1.4**

Let A be some $m \times n$ matrix.

Consider a list of k elementary row operations:

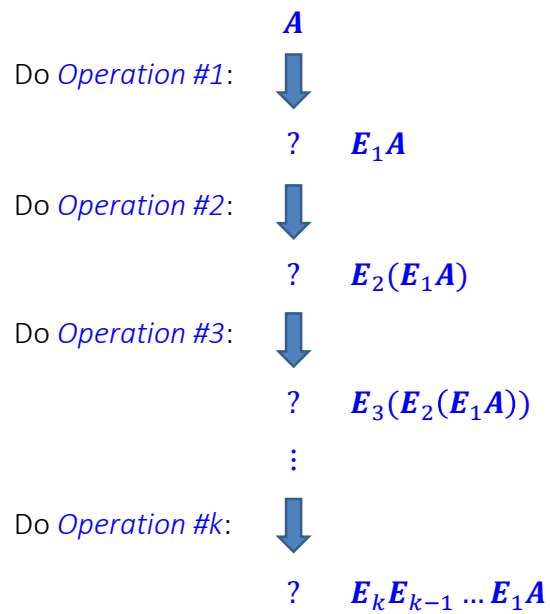
- *Row operation #1*, with corresponding elementary matrix E_1 .
- ...
- *Row operation #k*, with corresponding elementary matrix E_k .

Problem:

Write down an expression for the matrix that results from applying these operations in this order to the matrix A .

Solution:

Solution: We want to calculate the following sequence:



Video 6.1.5

So.

What is the reason the **Fundamental Fact** is true?

In other words, why do elementary matrices implement elementary row operations?

One way to see this is just by calculating the effect of these matrices directly.

This is fairly simple but does not give you much mathematical insight into why this fact should be true.

There is another, more conceptual way to see the underlying reason. The other way is to first understand the following:

Lemma (Row operations respect matrix multiplication):

Let $R(A)$ denote the result of applying some chosen row operation R to some matrix A . Then: $R(AB) = R(A)B$.

First let's check this lemma in a simple example.

We'll check it for the case:

- $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$
- $B = \begin{bmatrix} -1 & 0 & 1 \\ 2 & 0 & -1 \end{bmatrix}$
- The elementary row operation R given by $R_1 \rightarrow R_1 - 2R_2$

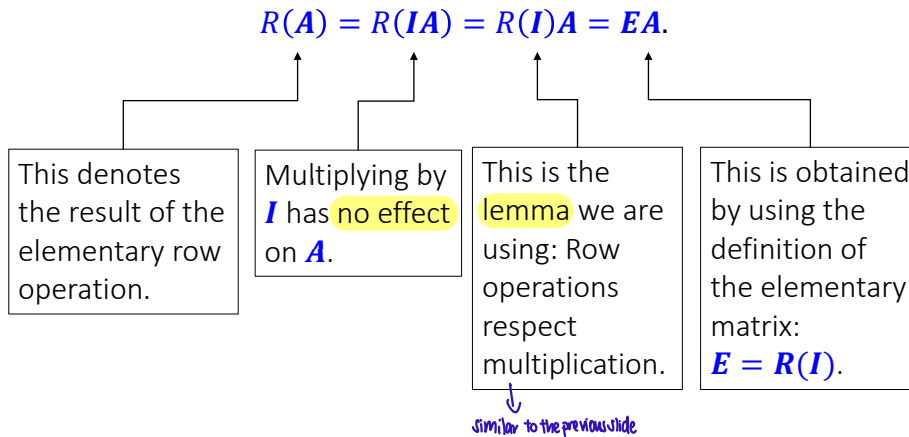
$R(AB) = R(A)B$
 LHS:
 $AB = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} -1 & 0 & 1 \\ 2 & 0 & -1 \end{bmatrix}$
 $= \begin{bmatrix} -1 & 0 & 1 \\ 5 & 0 & -1 \end{bmatrix}$
 $R(AB) = \begin{bmatrix} -3 & 0 & 1 \\ 5 & 0 & -1 \end{bmatrix}$
 RHS:
 $R(A) = \begin{bmatrix} -1 & -2 \\ 3 & 4 \end{bmatrix}$
 $R(A)B = \begin{bmatrix} -1 & -2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} -1 & 0 & 1 \\ 2 & 0 & -1 \end{bmatrix}$
 $= \begin{bmatrix} -3 & 0 & 1 \\ 5 & 0 & -1 \end{bmatrix}$

Same answer

Video 6.1.6

Assuming this lemma, the proof of the Fundamental Fact about elementary matrices is extremely simple. Here it is:

Proof of the Fundamental Fact using the lemma:



Example of this underlying fact $R(AB) = R(A)B$.

Consider the special case of the row operation $R_i \rightarrow R_i + aR_j$.

Organise A into rows and B into columns:

$$A = \begin{bmatrix} \leftarrow & A_1 & \rightarrow \\ & \vdots & \\ \leftarrow & A_n & \rightarrow \end{bmatrix}, \quad B = \begin{bmatrix} \uparrow & & \uparrow \\ B_1 & \cdots & B_m \\ \downarrow & & \downarrow \end{bmatrix}.$$

$\downarrow_{n \times 1}$ $\downarrow_{1 \times m}$

Then AB equals:

$$AB = \begin{bmatrix} A_1 B_1 & \cdots & A_1 B_m \\ \vdots & \cdots & \vdots \\ A_n B_1 & \cdots & A_n B_m \end{bmatrix}.$$

The result of the row operation on this, $R(AB)$, equals:

$$R(AB) = \begin{bmatrix} A_1 B_1 & \cdots & A_1 B_m \\ \vdots & \cdots & \vdots \\ A_i B_1 + a A_j B_1 & \cdots & A_i B_m + a A_j B_m \\ \vdots & \cdots & \vdots \\ A_n B_1 & \cdots & A_n B_m \end{bmatrix}.$$

$\rightarrow (A_i + a A_j) B_m$

Rebracketting entries this equals:

$$= \begin{bmatrix} A_1 B_1 & \cdots & A_1 B_m \\ \vdots & \cdots & \vdots \\ (A_i + aA_j) B_1 & \cdots & (A_i + aA_j) B_m \\ \vdots & \cdots & \vdots \\ A_n B_1 & \cdots & A_n B_m \end{bmatrix}.$$

And this equals a product of two matrices:

$$= \begin{bmatrix} A_1 \\ \vdots \\ A_i + aA_j \\ \vdots \\ A_n \end{bmatrix} [B_1 \quad \cdots \quad B_m] = R(A)B.$$

Similar elementary calculations can easily be performed for the other two elementary row operations.



Video 6.1.7

Elementary matrices are all invertible.

Quick reminder of the definition of invertible: A matrix A is invertible if there exists another matrix A^{-1} such that

$$AA^{-1} = A^{-1}A = I_n.$$

Fact

Every elementary matrix is invertible.

*The inverse of an elementary matrix is another elementary matrix.

Why?

We'll give two approaches to understanding this:

- A theoretical explanation: We'll see this is a consequence of the fact that all three elementary row operations on matrices can be *reversed* by other elementary row operations.
- And we can also easily write down the inverse matrices directly.

The key observation we'll use:

Elementary operations can always be undone by another elementary operation.

Example:

$$A \xrightleftharpoons{R_i \rightarrow R_i + cR_j} A'$$

Question: How to get back A from A' ?

Answer: $R_i \rightarrow R_i - cR_j$

$$\begin{bmatrix} 1 & 2 & 0 \\ -1 & 0 & 3 \\ 0 & 2 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 0 \\ -1 & 0 & 3 \\ 0 & 2 & 4 \end{bmatrix}$$

$R_3 \rightarrow R_3 + 5R_2$ $R_3 \rightarrow R_3 - 5R_2$

$$\begin{bmatrix} 1 & 2 & 0 \\ -1 & 0 & 3 \\ -5 & 2 & 19 \end{bmatrix}$$

The key observation we'll use:

Elementary operations can always be undone by another elementary operation.

Example (continued):

$$A \xrightleftharpoons{R_i \leftrightarrow R_j} A'$$

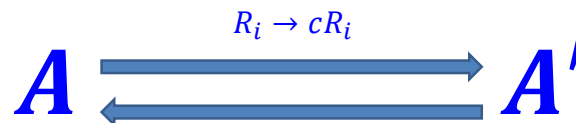
Question: How to get back A from A' ?

Answer: $R_i \leftrightarrow R_j$

The key observation we'll use:

Elementary operations can be undone by another elementary operation.

Example (continued):



Question: How to get back A from A' ?

Answer: $R_i \rightarrow \frac{1}{c}R_i$

Observation

Every elementary matrix is invertible.

Proof

Let E be an elementary matrix.

We need to explain why it has an inverse E^{-1} .

Well, if E is an elementary matrix then it corresponds to some elementary row operation R . Note that there will always be some other elementary row operation S which “undoes” the effect of the first row operation. (As we just discussed.)

Let F be the elementary matrix corresponding to the elementary row operation S that undoes R . (So $F = S(I)$.)

We'll check F is the inverse for E .

Proof (cont.)

Now just calculate $S(R(I_n))$ in two different ways:

$$\begin{array}{ccc}
 & S(R(I_n)) & \\
 \swarrow & & \searrow \\
 = S(EI_n) & & = (S \circ R)(I_n) \\
 = FEI_n & & = I_n. \\
 = FE. & &
 \end{array}$$

S is going to cancel R

Thus $FE = I_n$.

It is just as easy to prove $EF = I_n$.

Thus $F = E^{-1}$.

□

Observe an important detail in this proof:

Let E be an elementary matrix corresponding to some elementary row operation R .

The inverse of E is precisely the elementary matrix corresponding to the elementary row operation which “undoes” R .

Quick test of understanding

Video 6.1.8

Consider the elementary matrix $E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 7 & 1 \end{bmatrix}$.

What is its inverse?

Solution.

Above is the elementary matrix corresponding to the elementary row operation $R_3 \rightarrow R_3 + 7R_2$.

Note that the elementary row operation that undoes this is $R_3 \rightarrow R_3 - 7R_2$.

The elementary matrix corresponding to that operation is

$F = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -7 & 1 \end{bmatrix}$. This will be the inverse to E .

$$\text{Check: } \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -7 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 7 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -7 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 7 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Quick test of understanding

Consider the elementary matrix $E = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$.

What is its inverse?

Solution.

Above is the elementary matrix corresponding to the elementary row operation $R_1 \leftrightarrow R_3$.

Note that the elementary row operation that undoes this is the same operation $R_1 \leftrightarrow R_3$.

The elementary matrix corresponding to that operation is just E . So in this case E will be the inverse to itself.

$$\text{Check: } \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Quick test of understanding

Consider the elementary matrix $E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -100 & 0 \\ 0 & 0 & 1 \end{bmatrix}$.

What is its inverse?

Solution.

Above is the elementary matrix corresponding to the elementary row operation $R_2 \rightarrow (-100)R_2$.

Note that the elementary row operation that undoes this is the operation $R_2 \rightarrow (-\frac{1}{100})R_2$.

The corresponding elementary matrix $F = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -\frac{1}{100} & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Check: $\begin{bmatrix} 1 & 0 & 0 \\ 0 & -100 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & -\frac{1}{100} & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & -100 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Summary of examples of inverses

It is easy to write down an explicit inverse of an elementary matrix when we need one:

First type: $\left(\begin{bmatrix} 1 & 0 & a \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right)^{-1} = \begin{bmatrix} 1 & 0 & -a \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Second type: $\left(\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \right)^{-1} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$

Third type: $\left(\begin{bmatrix} 1 & 0 & 0 \\ 0 & c & 0 \\ 0 & 0 & 1 \end{bmatrix} \right)^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1/c & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Video 6.1.9

Representing sequences of elementary row operations by products of elementary matrices.

One of the most useful applications of elementary matrices is that we can use them to easily express sequences of elementary row operations *directly* as products of matrices.

Example:

The following two matrices are row equivalent.

$$\mathbf{A} = \begin{pmatrix} 1 & 0 & 2 \\ 2 & -1 & 3 \\ 1 & 4 & 4 \end{pmatrix} \quad \mathbf{B} = \begin{pmatrix} 1 & 0 & 2 \\ 0 & -1 & -1 \\ 0 & 0 & -2 \end{pmatrix}$$

We'll use this to find a sequence of elementary matrices $\mathbf{E}_i$ such that: $\mathbf{B} = \mathbf{E}_k \mathbf{E}_{k-1} \cdots \mathbf{E}_1 \mathbf{A}$. Step-by-step we replace an elementary row operation by its corresponding elementary matrix.

Example: We just do row operations as normal, but keep track of the corresponding matrices.

$$\mathbf{A} = \begin{pmatrix} 1 & 0 & 2 \\ 2 & -1 & 3 \\ 1 & 4 & 4 \end{pmatrix} \xrightarrow[\mathbf{E}_1]{R_2 - 2R_1} \begin{pmatrix} 1 & 0 & 2 \\ 0 & -1 & -1 \\ 1 & 4 & 4 \end{pmatrix}$$

$$\mathbf{E}_1 = \begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Example: We just do row operations as normal, but keep track of the corresponding matrices.

$$A = \begin{pmatrix} 1 & 0 & 2 \\ 2 & -1 & 3 \\ 1 & 4 & 4 \end{pmatrix} \xrightarrow[\mathbf{E}_1]{R_2 - 2R_1} \begin{pmatrix} 1 & 0 & 2 \\ 0 & -1 & -1 \\ 1 & 4 & 4 \end{pmatrix} \xrightarrow[\mathbf{E}_2]{R_3 - R_1} \begin{pmatrix} 1 & 0 & 2 \\ 0 & -1 & -1 \\ 0 & 4 & 2 \end{pmatrix}$$

$$\mathbf{E}_1 = \begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad \mathbf{E}_2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix}$$

Example: We just do row operations as normal, but keep track of the corresponding matrices.

$$A = \begin{pmatrix} 1 & 0 & 2 \\ 2 & -1 & 3 \\ 1 & 4 & 4 \end{pmatrix} \xrightarrow[\mathbf{E}_1]{R_2 - 2R_1} \begin{pmatrix} 1 & 0 & 2 \\ 0 & -1 & -1 \\ 1 & 4 & 4 \end{pmatrix} \xrightarrow[\mathbf{E}_2]{R_3 - R_1} \begin{pmatrix} 1 & 0 & 2 \\ 0 & -1 & -1 \\ 0 & 4 & 2 \end{pmatrix}$$

$$\xrightarrow[\mathbf{E}_3]{R_3 + 4R_2} \begin{pmatrix} 1 & 0 & 2 \\ 0 & -1 & -1 \\ 0 & 0 & -2 \end{pmatrix} = \mathbf{B}$$

$$\mathbf{E}_1 = \begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad \mathbf{E}_2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix} \quad \mathbf{E}_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 4 & 1 \end{pmatrix}$$

Conclusion: So $E_3 E_2 E_1 A = B$ where

$$E_1 = \begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad E_2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix} \quad E_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 4 & 1 \end{pmatrix}$$

Once we know this we can immediately write an analogous formula to go backwards from B to A :

$$A = (E_3 E_2 E_1)^{-1} B = E_1^{-1} E_2^{-1} E_3^{-1} B$$

where these inverses of elementary matrices can be written straight down:

$$E_1^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad E_2^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix} \quad E_3^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -4 & 1 \end{pmatrix}$$

Video 6.2.1

An important application of elementary matrices: characterizing invertibility.

Quick reminder:

" A is invertible" is defined to mean there exists another matrix A^{-1} such that

$$AA^{-1} = A^{-1}A = I.$$

For example: $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ is invertible because:

We can find a second matrix $\begin{bmatrix} -2 & 1 \\ 3/2 & -1/2 \end{bmatrix}$ with the property that

$$\begin{bmatrix} -2 & 1 \\ 3/2 & -1/2 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} -2 & 1 \\ 3/2 & -1/2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

An important application of elementary matrices: characterizing invertibility.

We'll use elementary matrices to give a few different characterisations of invertible matrices.

What do we mean by **characterization**?

Here we mean: we'll describe some simple properties of a matrix different to "invertible", and then prove that these new properties are true for a matrix exactly when that matrix is invertible.

Theorem. (Characterisation of Invertibility.)

Let A be a square $n \times n$ matrix.

The following four statements are equivalent:

1. The matrix A is invertible. ↗ Algebra
2. The homogeneous linear system $Ax = 0$ (where x is a $n \times 1$ matrix) only has the trivial solution (where every entry of x is zero). ↗ systems of equations
3. The reduced row-echelon form of A is the identity matrix I_n . ↗ row operations
4. The matrix A can be expressed as a product of elementary matrices. ↗ elementary matrices

↗ all these properties will be true when the matrix is invertible

Example: (From midterm 2016).

Video 6.2.2

Consider the following matrix which depends on 3 unspecified parameters a, b, c .

$$\begin{bmatrix} a & b & c \\ a & 0 & 0 \\ a & 0 & c \end{bmatrix}.$$

For what values of the parameters is this matrix invertible?

Solution

According to our characterization of invertibility #2 this matrix has an inverse exactly when the following matrix equation has a **unique solution**

$$\begin{bmatrix} a & b & c \\ a & 0 & 0 \\ a & 0 & c \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

Solution (continued).

So we need to find out when the following equivalent linear system has exactly 1 solution:

$$\left[\begin{array}{ccc|c} a & b & c & 0 \\ a & 0 & 0 & 0 \\ a & 0 & c & 0 \end{array} \right]$$

To find out this we just follow procedures we know already:

$$\left[\begin{array}{ccc|c} a & b & c & 0 \\ a & 0 & 0 & 0 \\ a & 0 & c & 0 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} a & b & c & 0 \\ 0 & -b & -c & 0 \\ 0 & -b & c & 0 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} a & b & c & 0 \\ 0 & -b & -c & 0 \\ 0 & 0 & c & 0 \end{array} \right]$$

This consistent system will have a leading entry in every column, hence exactly 1 solution, precisely when $a \neq 0, b \neq 0, c \neq 0$. We deduce that the given matrix A is invertible if and only if $a \neq 0, b \neq 0, c \neq 0$.

□

Video 6.2.3

Theorem. (Characterisation of Invertibility.)

Let A be a square $n \times n$ matrix.

The following four statements are equivalent:

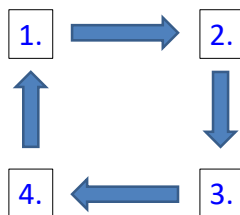
1. The matrix A is invertible.
2. The homogeneous linear system $Ax = 0$ (where x is a $n \times 1$ matrix) only has the trivial solution (where every entry of x is zero).
3. The reduced row-echelon form of A is the identity matrix I_n .
4. The matrix A can be expressed as a product of elementary matrices.

Proof:

We'll prove this in 4 steps:

1. Property 1 implies property 2.
2. Property 2 implies property 3.
3. Property 3 implies property 4.
4. Property 4 implies property 1.

We can represent this diagrammatically as follows:



For example, this gives $1 \Leftrightarrow 4$.

One direction $4 \Rightarrow 1$ is explicit.

The other direction $1 \Rightarrow 4$ follows because $1 \Rightarrow 2 \Rightarrow 3 \Rightarrow 4$.

Step 1. (1 $\Rightarrow$ 2)

We are assuming: A is invertible. This means the inverse A^{-1} exists. (Denote this fact $(\star)$.)

We must show it follows from this that: $Ax = 0$ only has a trivial solution.

So let x be any $m \times 1$ column vector solving the equation $Ax = 0$. (Denote this equation $(\star\star)$.)

Then:

$$\begin{aligned} x &= \overset{I}{\underbrace{A^{-1}A}}x \quad (\text{Using } (\star).) \\ &= A^{-1}0 \quad (\text{Using } (\star\star).) \\ &= 0. \quad (\text{Using a basic property of a zero matrix.}) \end{aligned}$$

Thus: the only solution is 0 as required.

Step 2. (2 $\Rightarrow$ 3)

Video 6.2.4

We are assuming: $Ax = 0$ only has the trivial solution $x = 0$.

We must show it follows from this that: the reduced row echelon form of A is the identity matrix I_n .

Start by forming the augmented matrix $(A|0)$ corresponding to the system $Ax = 0$. Then apply Gauss-Jordan elimination to put this matrix into reduced row echelon form.

Note that every column in RREF must contain a leading entry.

Why?

If there is a column without a leading entry in RREF form, then that column becomes a free parameter for the set of solutions of $Ax = 0$. And if there is a free parameter then there are infinitely many solutions. But we are assuming there is only one, 0 .

Step 2 (cont.):

identity matrices are square matrices

A square matrix in RREF form where every column contains a leading entry must be the identity matrix, as required.

Step 3. ($3 \Rightarrow 4$)

Video 6.2.5

We are assuming: the reduced row echelon form of A is the identity matrix I_n .

We must show it follows from this that: A can be expressed as a product of elementary matrices.

The assumption implies that A can be transformed into I_n by a sequence of elementary row operations. And thus, by just reversing the operations, I_n can be transformed into A by a sequence of elementary operations.

It follows that there are elementary matrices $E_1, \dots, E_k$ such that

$$A = E_k \dots E_1 I_n = E_k \dots E_1$$

as required.

Step 4. ($4 \Rightarrow 1$)

Video 6.2.6

We are assuming: the matrix A can be expressed as a product of elementary matrices.

We must show it follows from this that: A is invertible.

The assumption says that there are elementary matrices $E_1, \dots, E_k$ such that

$$A = E_1 \dots E_k.$$

Now observe:

- Elementary matrices are invertible.
- Any product of invertible matrices is invertible.
- Thus A is invertible, as required.

(Comment: An explicit formula for the inverse is of course

$$A^{-1} = E_k^{-1} \dots E_1^{-1}.$$

$$(E_1 E_2 \dots E_k)^{-1}$$

□

$$\begin{array}{c} A^{-1} A = I \\ \swarrow \quad \searrow \\ E_k^{-1} \dots E_2^{-1} E_1^{-1} \quad E_1 E_2 \dots E_k \end{array}$$

Video 6.3.1

An algorithm to find the inverse

If we think through the proof of the “Characterization of invertibility”, we can actually extract from it an explicit algorithm to compute the inverse matrix A^{-1} from an invertible matrix A .

First we’ll meet the algorithm, then we’ll think about why it works.

The algorithm:

Let $\mathbf{A}$ be an *invertible* $n \times n$ matrix.

Form a new matrix by putting the $n \times n$ identity matrix $\mathbf{I}_n$ to the right of $\mathbf{A}$. For clarity we usually draw a vertical line separating the two matrices. So we have written a matrix: $[\mathbf{A}|\mathbf{I}_n]$.

Now note that because we are assuming $\mathbf{A}$ is invertible we know from the “Characterising Invertibility” theorem that the reduced row-echelon form of $\mathbf{A}$ is the identity matrix $\mathbf{I}_n$. Thus, if we apply Gauss-Jordan elimination to $\mathbf{A}$ we will end up with $\mathbf{I}_n$.

The algorithm is:

- Do the same row operations you would do in Gauss-Jordan elimination to transform $\mathbf{A}$ into $\mathbf{I}_n$ but apply them to $[\mathbf{A}|\mathbf{I}_n]$.
- After this $[\mathbf{A}|\mathbf{I}_n]$ is transformed into $[\mathbf{I}_n|\mathbf{X}]$ for some $\mathbf{X}$.
- It turns out that $\mathbf{X}$ is the inverse of $\mathbf{A}$.

Example

Find the inverse of

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{pmatrix}$$

if it exists.

Example (cont.)

$$\begin{aligned}
 & \begin{pmatrix} 1 & 2 & 3 & | & 1 & 0 & 0 \\ 2 & 5 & 3 & | & 0 & 1 & 0 \\ 1 & 0 & 8 & | & 0 & 0 & 1 \end{pmatrix} \xrightarrow{\substack{R_2 - 2R_1 \\ R_3 - R_1}} \begin{pmatrix} 1 & 2 & 3 & | & 1 & 0 & 0 \\ 0 & 1 & -3 & | & -2 & 1 & 0 \\ 0 & -2 & 5 & | & -1 & 0 & 1 \end{pmatrix} \xrightarrow{R_3 + 2R_2} \\
 & \begin{pmatrix} 1 & 2 & 3 & | & 1 & 0 & 0 \\ 0 & 1 & -3 & | & -2 & 1 & 0 \\ 0 & 0 & -1 & | & -5 & 2 & 1 \end{pmatrix} \xrightarrow{-R_3} \begin{pmatrix} 1 & 2 & 3 & | & 1 & 0 & 0 \\ 0 & 1 & -3 & | & -2 & 1 & 0 \\ 0 & 0 & 1 & | & 5 & -2 & -1 \end{pmatrix} \xrightarrow{\substack{R_1 - 3R_3 \\ R_2 + 3R_3}} \\
 & \begin{pmatrix} 1 & 2 & 0 & | & -14 & 6 & 3 \\ 0 & 1 & 0 & | & 13 & -5 & -3 \\ 0 & 0 & 1 & | & 5 & -2 & -1 \end{pmatrix} \xrightarrow{R_1 - 2R_2} \begin{pmatrix} 1 & 0 & 0 & | & -40 & 16 & 9 \\ 0 & 1 & 0 & | & 13 & -5 & -3 \\ 0 & 0 & 1 & | & 5 & -2 & -1 \end{pmatrix}
 \end{aligned}$$

A^{-1}

Example (cont.)

$$\begin{pmatrix} 1 & 2 & 3 & | & 1 & 0 & 0 \\ 2 & 5 & 3 & | & 0 & 1 & 0 \\ 1 & 0 & 8 & | & 0 & 0 & 1 \end{pmatrix} \xrightarrow[\text{Elimination}]{\text{Gauss-Jordan}} \begin{pmatrix} 1 & 0 & 0 & | & -40 & 16 & 9 \\ 0 & 1 & 0 & | & 13 & -5 & -3 \\ 0 & 0 & 1 & | & 5 & -2 & -1 \end{pmatrix}$$

Interpretation:

First note that we have just shown that reduced row-echelon form on this matrix is I_3 . Thus, by the theorem “Characterising invertibility” this implies that this A is indeed invertible.

Moreover, the inverse matrix is the matrix we ended up with on the right hand side:

$$A^{-1} = \begin{bmatrix} -40 & 16 & 9 \\ 13 & -5 & -3 \\ 5 & -2 & -1 \end{bmatrix}. \quad \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix} \begin{bmatrix} -40 & 16 & 9 \\ 13 & -5 & -3 \\ 5 & -2 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Exercise**Video 6.3.2**

Use this method to find the inverse of the matrix

$$\begin{bmatrix} -1 & 0 & 1 \\ 0 & 1 & 0 \\ -2 & 5 & 1 \end{bmatrix}.$$

Solution

Following the procedure we write down:

$$\left[\begin{array}{ccc|ccc} -1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ -2 & 5 & 1 & 0 & 0 & 1 \end{array} \right]$$

Now we do Gauss-Jordan Elimination:

$$\begin{bmatrix} -1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ -2 & 5 & 1 & 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} -1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 5 & -1 & -2 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} -1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & -1 & -2 & -5 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} -1 & 0 & 0 & -1 & -5 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & -1 & -2 & -5 & 1 \end{bmatrix}$$

$$\downarrow$$

$$\begin{bmatrix} 1 & 0 & 0 & 1 & 5 & -1 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 2 & 5 & -1 \end{bmatrix} \leftarrow \begin{bmatrix} 1 & 0 & 0 & 1 & 5 & -1 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & -1 & -2 & -5 & 1 \end{bmatrix}$$

Solution (cont.)

Conclusion:

The inverse is: $\begin{bmatrix} 1 & 5 & -1 \\ 0 & 1 & 0 \\ 2 & 5 & -1 \end{bmatrix}$

Why does this algorithm work?

Video 6.3.3

Because A is invertible Gauss-Jordan elimination gives us some sequence of elementary row operations that transforms A into I_n .

Denote the elementary matrices corresponding to this sequence of operations by $E_1, \dots, E_k$.

Note that they satisfy the equation: $E_k \dots E_1 A = I_n$ $(\star)$.

If we apply this sequence of elementary row operations to the matrix $[A|I_n]$ then we obtain:

$$[A|I_n] \xrightarrow{\text{sequence of ops}} [E_k \dots E_1 A | E_k \dots E_1 I_n] = [I_n | E_k \dots E_1].$$

This shows that the matrix that appears on the right after this sequence of operations is precisely $E_k \dots E_1$.

According to equation $(\star)$ this is exactly A^{-1} . Also it is easy to see from this that the other equation $A E_k \dots E_1 = I_n$ follows as well.